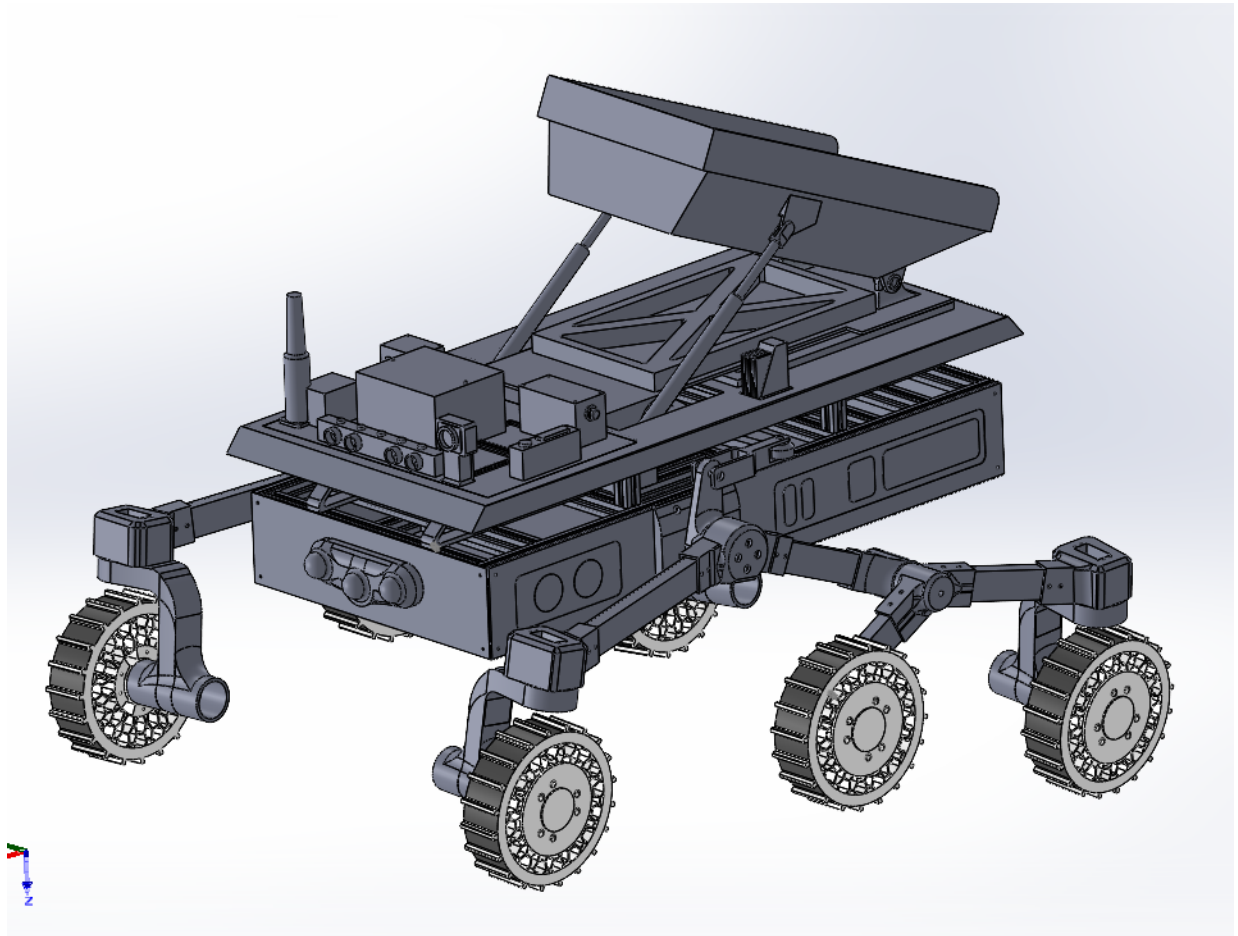


Week 12 Deliverable

Electronics, Communications, Sensor, Lighting, and Harness Integration

Conceptual Lunar Regolith Transport Rover



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This report documents the final Week 12 CAD arrangement and a complete preliminary electronics architecture. It distinguishes geometry visible in the supplied SOLIDWORKS screenshots from COTS packaging candidates, numerical design allocations, and unresolved mission requirements. The work is an integration study, not a flight-qualified electrical, thermal, radiation, or communications design.

Contents

1	Executive Summary	1
2	Week 11 Structural Baseline	1
3	Final CAD Integration Arrangement	2
4	Electronics Architecture	2
4.1	Required, optional, and mission-dependent equipment	2
5	Preliminary Component and Interface Table	6
6	Communications and Antenna Integration	9
6.1	Antenna CAD checks	9
7	Navigation, Perception, and Lighting	9
7.1	Forward stereo cameras	9
7.2	Ranging and hazard coverage	9
7.3	IMU and mechanism sensors	10
7.4	Lighting	10
8	Mechanical Mounting and Serviceability	10
8.1	Enclosures and brackets	10
8.2	Proposed internal enclosure assignment	10
9	Harness Routing Strategy	10
10	Preliminary Power and Energy Budget	11
11	Mass, Center of Mass, and Mounting Loads	12
11.1	Current mass evidence	12
11.2	Mounting-load equations	14
12	Thermal, Vacuum, Radiation, and Dust Considerations	14
13	Interference and Range-of-Motion Verification	15
14	Need for Updated Structural Analysis	15
15	Requirements Assessment	16
16	Risks, Limitations, and Recommended Future Work	16
17	Conclusion	18
18	References	19

1 Executive Summary

Week 12 integrated the rover's communications, computing, sensing, lighting, power-distribution, and cable-management functions onto the fixed forward upper deck. The final CAD places a central electronics enclosure, adjacent power and actuator-control housings, a roughly 400 mm antenna mast, a forward optical sensor, a multi-aperture front bar, an accessible service/status housing, and protected cable channels ahead of the dump mechanism. The long 1000 mm edge of the preliminary deck zone is the rover-front edge. The raised-bed screenshot shows no visible overlap between the dump bed or actuator structure and the populated forward deck.

The arrangement is mechanically credible because the electronics are on a fixed structural deck rather than the payload floor, heavy enclosures remain low, connectors face accessible perimeter edges, and the mast has a direct base on the deck. The central enclosure should contain the main computer, deterministic vehicle controller, Ethernet switch, data storage, and the rigidly mounted IMU. The two larger adjacent housings should be assigned to protected power distribution and the dual actuator drivers. The mast-base housing should contain the radio front end only if RF cable loss and thermal requirements support that choice.

Four manufacturer-traceable COTS devices are used strictly as packaging and power candidates: an NVIDIA Jetson AGX Orin-class compute platform, two Basler ace 2 GigE cameras, a Livox Mid-360 LiDAR, and a VectorNav VN-100 rugged IMU. Their sourced subtotal is 0.490 kg excluding the computer and all brackets, enclosures, connectors, and harnesses. At a 30 W compute mode, the candidate compute/camera/LiDAR/IMU set draws 43.52 W continuously and 81.02 W at the defined candidate peak. With a 20% allocation margin, its continuous energy is 0.177 kWh over the Week 9 3.395 h trip. This is not the rover electrical total: radio transmit power, work lights, actuator motors, heaters, wheel drives, and conversion losses remain unselected.

The current SOLIDWORKS Default-configuration mass-properties snapshot reports 423.85468 kg, a modeled solid volume of $0.42205884146 \text{ m}^3$, a surface area of 61.98119843 m^2 , and center-of-mass coordinates (142.88, 1629.41, 1829.83) mm in the displayed default coordinate system. This is the current Week 12 CAD assembly snapshot; it does not replace the separate mission estimates of approximately 650 kg dry and 990 kg fully loaded.

The final integration assessment is **preliminary pass with open verification items**. Placement, protection, service access, forward visibility, and capture of the current CAD mass properties pass. Full-motion clearance, antenna keep-out, stereo baseline, thermal rejection, radiation suitability, harness bend radius, bracket fasteners, and audit of all material/mass assignments remain open.

2 Week 11 Structural Baseline

The Week 11 report analyzed five linear-static cases using 6061-T6 aluminum with $E \approx 69 \text{ GPa}$, $\nu = 0.33$, density 2700 kg.m^{-3} , and parent-material yield strength 275 MPa. The 340 kg payload produces 550.8 N on the Moon. Dividing by the measured loaded area, 0.47069195 m^2 , produced the nominal pressure of approximately 1170 N.m^{-2} .

LC5 governed the completed set with 5.338 MPa maximum von Mises stress and 1.941% utilization of the 275 MPa parent-material yield strength. All completed cases passed the preliminary elastic strength and stiffness screen. The separate 3207.6 N full-rover lunar design-load case was **not completed** and is not presented as an analyzed result. Likewise, the early 74.2083 kg mass-properties snapshot excluded hidden bodies and shall not be used as the final rover or Week 12 integration mass.

The Week 11 margins indicate that low-mass electronics are unlikely to reverse the global parent-material conclusion. However, local mast-base, enclosure-foot, and fastener loads were not represented and must be checked after actual masses and hole patterns are known.

Table 1: Completed Week 11 global FEA results.

Case	Condition	Stress (MPa)	Displacement (mm)	FOS
LC1	Horizontal, nominal payload	0.072929	0.0008055	3770.8
LC2	Horizontal, doubled payload pressure	0.1002	0.0008055	2744.5
LC3	Raised, 25% retained payload	0.8681	0.01503	316.8
LC4	Raised, full retained/jammed payload	2.137	0.02220	128.7
LC5	Horizontal, asymmetric terrain support	5.338	0.1415	51.52

3 Final CAD Integration Arrangement

Figure 2 numbers the visible deck elements. The labels describe intended functional assignments; they are not claims that final internal hardware has been selected.

Table 2: Placement assessment from the supplied screenshots.

Criterion	Assessment	Status
Fixed versus moving structure	Major enclosures and mast are on the fixed forward deck, not the dump bed	PASS
Payload/dump-path protection	Equipment is forward of the bed and below the payload lip in the shown raised state	PASS
Mechanism interference	No overlap is visible at the shown raised position; continuous swept-volume evidence is absent	OPEN
Forward sensor visibility	Forward apertures and pedestal are not blocked by the raised bed in the screenshot	PASS
Antenna visibility	Mast is ahead of the bed; rear-hemisphere shadowing and RF keep-out are not yet modeled	OPEN
Service access	Top covers and perimeter connectors appear accessible with the bed raised	PASS
Thermal rejection	Low, separated enclosures offer deck conduction area; radiator path and coating are undefined	OPEN
Structural attachment	Mast and boxes visibly contact the deck, but fasteners, doublers, and deck load paths are not documented	OPEN

4 Electronics Architecture

The architecture separates high-level autonomy from deterministic mechanism safety. The main computer processes cameras and ranging data, while a dedicated vehicle controller executes limit-switch logic, watchdog functions, actuator interlocks, and safe shutdown. Loss of the high-level computer shall not permit uncommanded bed motion.

4.1 Required, optional, and mission-dependent equipment

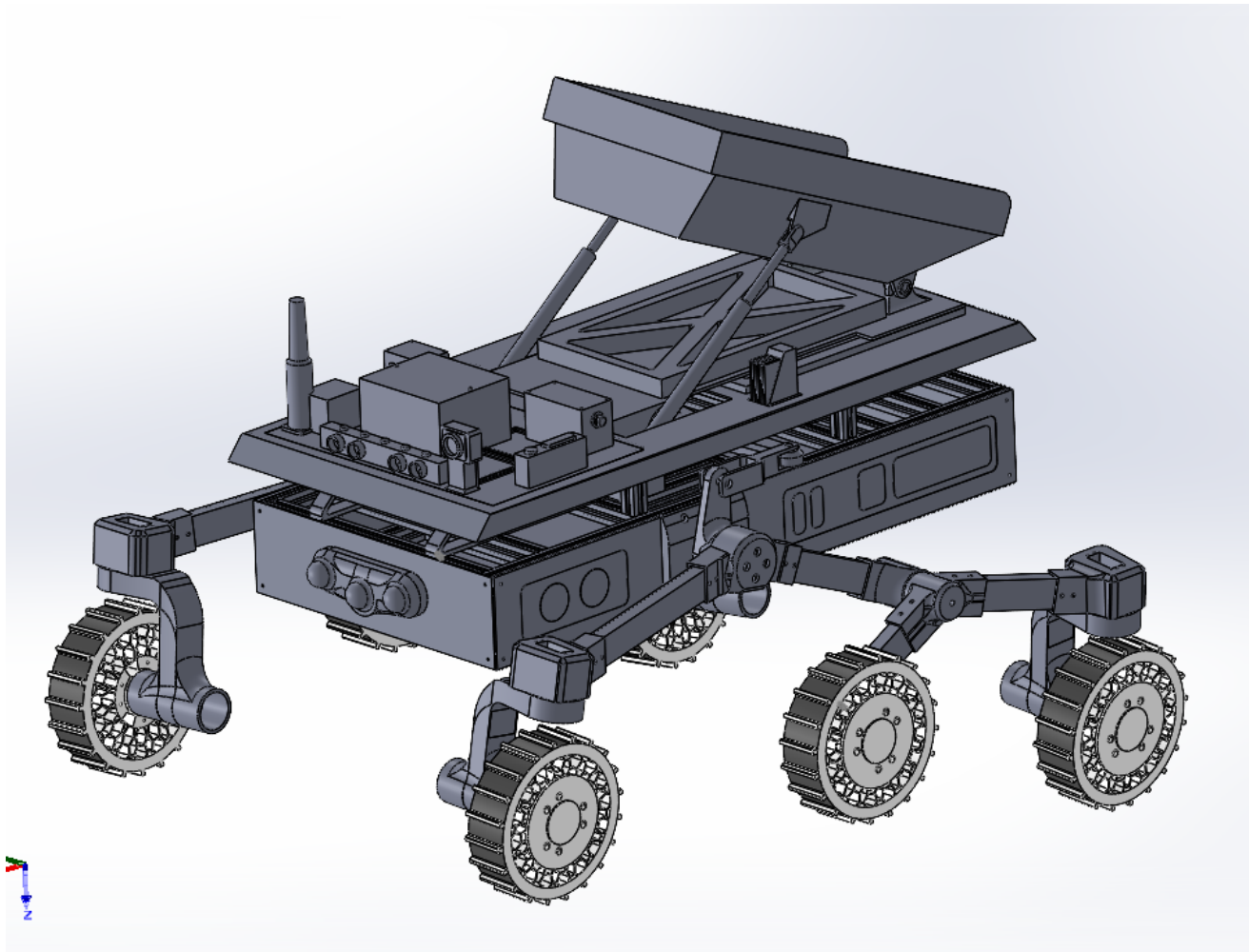


Figure 1: Final Week 12 rover CAD with the dump bed raised. Electronics are concentrated on the fixed forward upper deck. The screenshot verifies the intended overall arrangement but not swept-volume clearance.

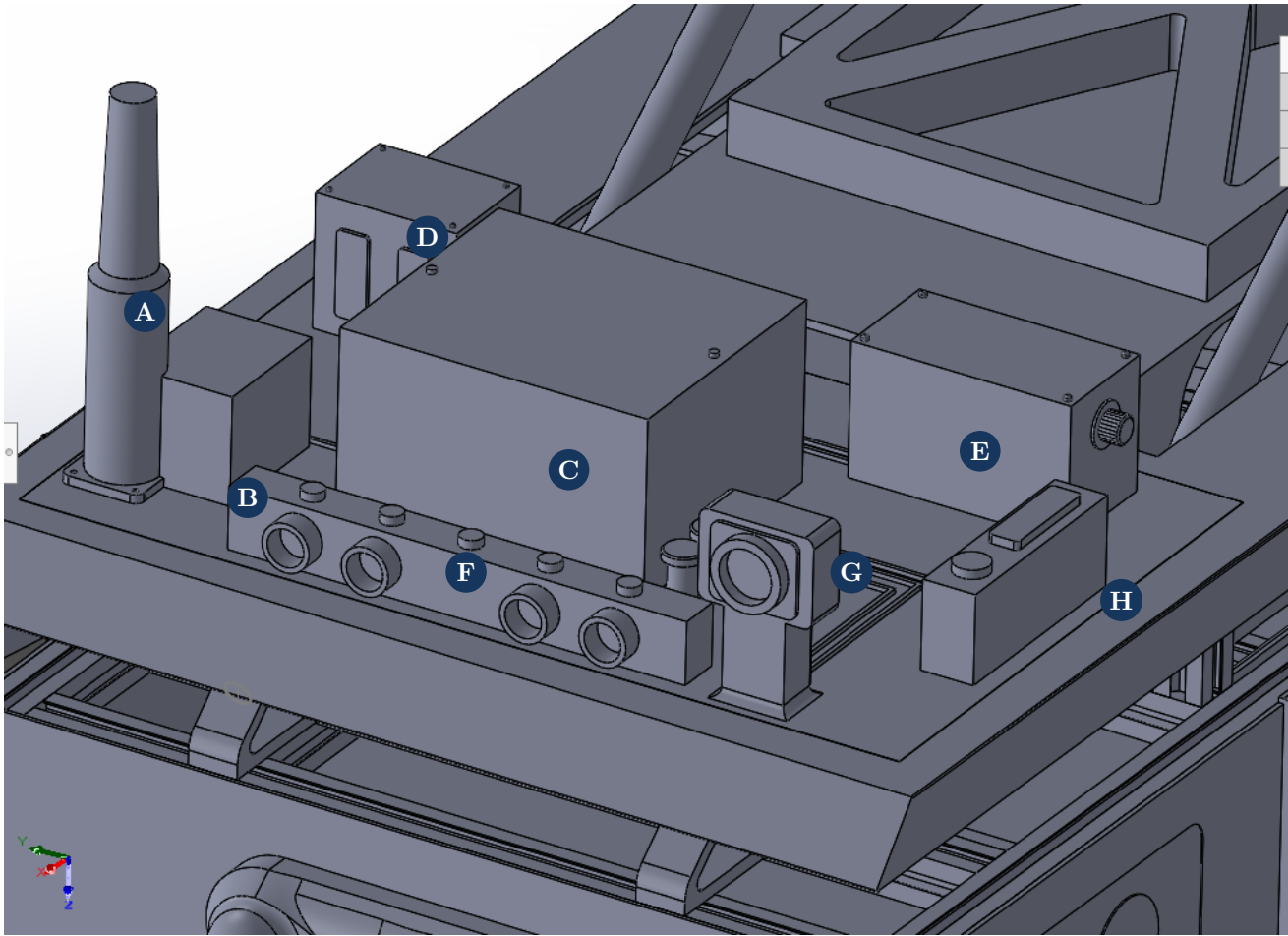


Figure 2: Close-up of the final forward electronics deck. A: antenna/mast placeholder; B: mast-base radio or RF interface box; C: main electronics enclosure; D: power-distribution/protection enclosure; E: dual actuator-driver enclosure; F: multi-aperture work-light and short-range-sensor bar; G: forward camera or ranging-sensor pedestal; H: service disconnect/status interface.

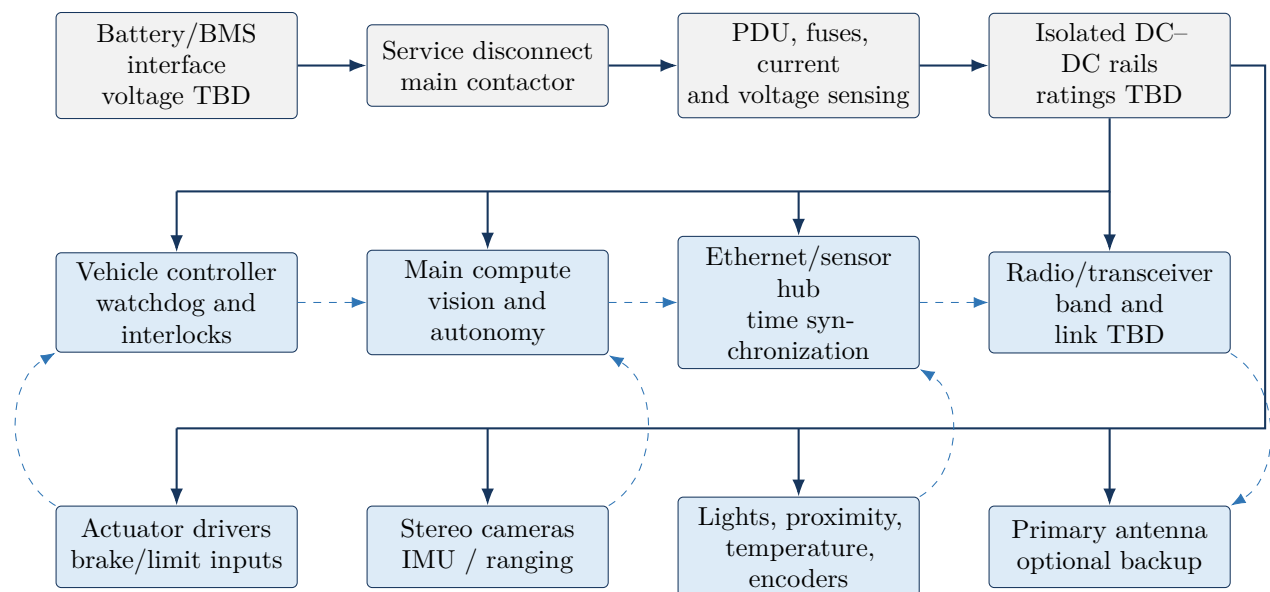


Figure 3: Preliminary electrical and data architecture. Solid arrows show power distribution; dashed arrows show principal command or data paths. Traction power is outside the detailed Week 12 deck scope.

Table 3: Complete preliminary equipment classification.

Category	Components and reason	Status
Computing and safety	Main autonomy computer; independent deterministic vehicle controller; hardware watchdog; nonvolatile data storage; Ethernet/sensor switch	Required
Power interface	Battery/BMS interface; service disconnect; contactor/precharge as required; PDU; branch fuses or electronic protection; current/voltage monitoring; required DC–DC converters; chassis bonding point	Required
Dump mechanism	Two-channel actuator driver; actuator stroke feedback or position transducers; fully lowered and fully raised limit switches; hinge/bed-angle sensor; actuator and driver temperature sensors	Required
Navigation	Forward synchronized stereo-camera pair; rigid-body IMU/inclinometer; six wheel-speed/position interfaces; time synchronization	Required
Hazard detection	Front and rear/side hazard coverage using cameras or short-range ranging sensors; dump-zone/pinch-zone awareness	Required
Communications	Primary transceiver, antenna, RF feed, protected mast base, command-loss behavior	Required ; exact RF design is Mission decision
Lighting/indication	Sealed forward work-light assembly; visible status/fault indicator; fused/controlled lighting branch	Required
Harness hardware	Space-compatible wire and jacket selection; shield termination; sealed connectors/glands; strain relief; clamps; abrasion sleeve; removable deck and actuator harness sections; flexible service loops	Required
Thermal/environment	Enclosure and actuator temperature sensing; conductive heat spreaders/interface pads; radiator or controlled rejection surface; dust seals/covers; grounding/ESD control	Required
LiDAR/depth sensor	Adds range geometry and hazard-detection redundancy. Use only if field of view, dust response, processing load, and qualification justify it	<i>Optional</i>
Secondary antenna/radio path	Provides geometric or hardware redundancy; justified only by link-availability and fault-tolerance requirements	<i>Optional</i> / Mission decision
Rear/side work lights	Improve maintenance and reverse driving, but add power and dust-exposed optics	<i>Optional</i>
Payload load cells	Can estimate retained payload but require calibrated structural load paths and thermal compensation	<i>Optional</i>

Category	Components and reason	Status
Lunar PNT receiver	Future LunaNet/Moonlight-compatible receiver if the mission and service availability require it; no terrestrial GPS assumption	Mission decision
Heaters	Survival and operational heaters may dominate the budget; sizing requires landing site, duty cycle, shadow/night exposure, insulation, radiator, and allowable temperatures	Mission decision

5 Preliminary Component and Interface Table

Table 4 uses actual manufacturer data for four sizing candidates. These parts are not represented as lunar-qualified. All other rows are functional requirements whose numerical values remain deliberately open until the actual actuator, battery, radio link, thermal case, and connector family are selected.

Table 4: Manufacturer-traceable packaging candidates.

Function and candidate	Quantity, envelope, and mass	Power	Voltage and data interface	Proposed location, source, and qualification status
Autonomy compute: NVIDIA Jetson AGX Orin developer-kit envelope / Orin-class production module	Qty. 1; $110 \times 110 \times 71.65$ mm developer-kit envelope; mass TBD	15–60 W configurable	Ethernet, USB and CAN/GPIO available; production carrier input is TBD	Inside central enclosure C on a conductive base. NVIDIA [5]. Candidate only; no lunar qualification claimed.
Stereo navigation: Basler ace 2 a2A1920-51gcPRO global-shutter cameras	Qty. 2; $55.5 \times 29 \times 29$ mm and 0.105 kg each, excluding lens/connector clearance	3.4 W typical each	PoE or 12–24 VDC; GigE data and M8 I/O	Rigid matched pair across the forward edge. Basler [6]. IP30 terrestrial product; not lunar rated.
Three-dimensional ranging: Livox Mid-360 LiDAR	Qty. 1; $65 \times 65 \times 60$ mm; 0.265 kg	6.5 W average; 14 W cold self-heating peak	9–27 VDC; 100BASE-TX Ethernet	Raised front pedestal or dedicated mast collar after occlusion check. Livox [7]. Optional terrestrial candidate.
Rigid-body attitude: VectorNav VN-100 rugged IMU/AHRS	Qty. 1; $36 \times 33 \times 9$ mm; 0.015 kg	0.22 W	4.5–5.5 V; serial TTL or RS-232	Inside C on the chassis datum, away from high-current/magnetic noise. VectorNav [8]. Candidate only.

Table 5: Required functions with selections still open.

Function	Quantity	Proposed location	Inputs required before selection
Safety controller and watchdog	1 set	Central enclosure C, electrically separated from the main compute	I/O count, redundancy, radiation and thermal requirements
Radio/transceiver	1	B for a short RF feed, or C if thermal/service constraints dominate	RF band, range, relay/Earth geometry, data rate, outage tolerance and link margin
Primary antenna	1	Approximately 400 mm mast A with an independent deck load path	Antenna type, polarization, gain, ground plane, feed loss and keep-out
Power distribution and protection	1 set	Enclosure D near the deck power entry	Battery bus, fault current, contactor/precharge need, branch loads and grounding scheme
DC–DC converters	As required	D or C on a controlled conductive heat path	Input range, output rails, isolation, efficiency, derating and thermal case
Actuator drivers	2 channels	Enclosure E near the fixed actuator harness branch	Actuator voltage, stall/rated current, braking, duty cycle, feedback and command interface
Bed-angle and limit sensing	1 angle sensor plus 2 end switches	Hinge axis and protected fixed stops	Required accuracy, redundancy, switch geometry and temperature range
Actuator position sensing	2	Fixed actuator bodies with controlled service loops as needed	Stroke, feedback type, accuracy, mounting and environmental qualification
Hazard proximity	4–6 preliminary	Front bar F and protected rear/side corners	Coverage, range, FOV, dust response, interface and redundancy
Forward work light	1 minimum	Existing through-mounted front chassis location	Lamp envelope, power, beam pattern, fasteners, sealing and thermal path
Status/service interface	1 set	Housing H on the accessible forward perimeter	Disconnect rating, connector family, keying, indication and fault behavior
Temperature monitoring	6 minimum	Compute, PDU, driver, radio, both actuators and thermal reference points	Sensor type, accuracy, trip limits, harness and controller inputs
Harness and connectors	1 routed set	Perimeter channels, fixed-deck clamps and removable moving branches	Branch currents, gauge, length, bend radius, shielding, sealing and connector family

6 Communications and Antenna Integration

The approximately 400 mm mast is a reasonable mechanical starting height because it raises the antenna above the low enclosures without creating a very tall rover appendage. It is not an RF-validated height. Final antenna type, ground plane, mast material, feed loss, pointing, polarization, data rate, transmit power, coding margin, Earth/relay geometry, and permissible outage duration require a formal link budget.

The mast should be attached through a base plate or short tube socket that transfers moment into a deck rail or local doubler; it should not rely on the 1 mm layout sheet. The antenna radiator should remain above nearby metallic lids by the keep-out distance specified in the selected antenna datasheet. If a radio enclosure is placed at the mast base, the electrical design must trade reduced RF feed loss against additional solar/dust exposure and the need to carry power and high-speed data to the box.

NASA describes LunaNet as a future interoperable communications and navigation architecture for lunar surface and orbital users [9]. A NASA lunar architecture study also notes that conventional GPS use at lunar distance is limited by weak signals, poor geometry, and Earth occultation [10]. Therefore this rover's baseline navigation uses visual odometry, wheel odometry, inertial sensing, and ranging. A LunaNet-compatible PNT receiver is a mission option, not an assumed GPS unit.

6.1 Antenna CAD checks

1. Create a transparent cylindrical or manufacturer-defined antenna keep-out body concentric with the radiator.
2. Evaluate it in horizontal-bed and raised-bed configurations and at intermediate actuator strokes.
3. Create line-of-sight rays toward the intended surface relay, lunar satellite, or Earth direction set. Record bed and enclosure occlusion angles.
4. Run interference detection between the mast assembly, bed swept volume, actuator envelopes, service covers, and handling/removal paths.
5. After mass is selected, check mast base bending and fasteners using Section 11.2.

7 Navigation, Perception, and Lighting

7.1 Forward stereo cameras

The final CAD visibly contains one square forward optical unit. Stereo navigation requires either a second matched camera or replacement of that single placeholder with an integrated stereo head. The two optical centers must be rigidly separated by a measured baseline, share synchronized exposure, and retain overlapping forward fields of view. Lens focal length and horizontal/vertical FOV are not selected; therefore no numerical range-resolution or coverage claim is made.

The camera windows should sit ahead of enclosure edges and below a small dust/impact brow without vignetting. A removable transparent cover, purge-free dust mitigation strategy, or electrodynamic cleaning concept should be evaluated. NASA identifies lunar dust as abrasive and electrostatic, capable of clinging to and damaging hardware [11].

7.2 Ranging and hazard coverage

The Mid-360 is an optional packaging candidate because it supplies 360° horizontal by 59° vertical coverage, a 0.1 m close-proximity blind zone, and a 40 m stated range at 10% reflectivity [7]. Its terrestrial rating does not establish performance in lunar vacuum, radiation, or dust. If it is mounted on the current low forward pedestal, the central enclosure may block a large rear sector. A dedicated raised pedestal or a mechanically separate collar below the antenna should be compared before selecting it.

At minimum, the rover needs forward and rear/side hazard coverage during driving, plus awareness of the dump-zone and actuator pinch regions during bed operation. The multi-aperture front bar may house work-light optics and short-range sensors, but the assignment of each aperture must be fixed in the BOM and annotated in CAD before release.

7.3 IMU and mechanism sensors

The IMU should be installed inside the central enclosure on a machined or stiff bracket tied directly to the chassis datum. Its axes and lever-arm vector to the chosen rover reference point must be documented. It should be separated from high-current driver loops and ferromagnetic hardware; magnetic heading should not be assumed reliable without site calibration.

The dump-bed safety chain requires: two actuator position measurements, a bed-angle measurement independent of inferred actuator stroke, a fully lowered switch, a fully raised switch, actuator overcurrent detection, and temperature sensors on both actuators and the driver enclosure. Software limits are useful but do not replace independent end-of-travel inputs.

7.4 Lighting

The current rover CAD includes a three-element front work-light assembly through-mounted into the chassis. This is a good protected location because it stays on the fixed frame and remains below the raised bed. The final hardware should use a sealed optical assembly, a concealed cable pass-through, strain relief inside the chassis, a fused lighting output, and through-bolts with nutplates/inserts if the wall is too thin for durable threads. Exact wall thickness, fastener size, lamp power, beam angle, and photometric performance remain unconfirmed.

8 Mechanical Mounting and Serviceability

8.1 Enclosures and brackets

All large housings should use bottom flanges or simple angle brackets with accessible top-side fasteners. The deck attachment should include at least one anti-rotation feature and a defined chassis-bonding path. Fastener type, hole diameter, bracket thickness, fillet radius, edge distance, and thread engagement shall be selected only after the actual enclosure masses, base footprints, and deck/rail thicknesses are measured.

The main enclosure lid should be removable without disturbing the mast or the dump-bed mechanism. Connectors should face the front or outboard sides; rear-facing connectors near the mechanism are discouraged. Maintain tool clearance for each lid screw, connector coupling ring, fuse replacement, and service-disconnect operation.

8.2 Proposed internal enclosure assignment

9 Harness Routing Strategy

The harness should be modeled as separate fixed-deck, chassis, suspension, and dump-mechanism branches. Fixed runs should follow the deck perimeter or dedicated channels with regularly spaced clamps; routing across removable covers should be avoided. A separate removable connector should isolate the complete electronics-deck assembly.

- Route actuator motor power and lighting power separately from camera Ethernet, IMU, encoder, limit-switch, and RF cables where practical. Cross unavoidable power/signal intersections approximately orthogonally.

Table 6: Functional assignment of the modeled deck housings.

Label	Primary contents	Integration notes
B	Radio/front-end or protected RF junction	Use only after feed-loss and thermal trade; keep serviceable at mast base
C	Main compute, safety controller, storage, Ethernet switch, IMU	Conductive base, EMI partitioning, removable lid, internal strain relief
D	PDU, branch protection, contactor/precharge, primary DC–DC converters	Place near power entry; isolate noisy converter loops from IMU and RF
E	Two actuator drivers and braking/current sensing	Short fixed-frame motor leads; separate motor and sensor connectors
F	Work lights and/or short-range sensor apertures	Freeze aperture function and window materials; protect from dust and impact
G	Forward camera, stereo head, or ranging sensor	Add second camera if G is a single monocular camera
H	Service disconnect, status indicator, maintenance connector	Keyed and reachable; prevent accidental actuation

- Provide positive strain relief at every connector and gland. NASA-STD-8739.4 requires stress relief at connector exits and defines installed harness bend-radius criteria; for an overall harness containing coax or AWG 8 or larger conductors, the cited minimum is six times the outer diameter [12]. The selected cable datasheet governs if it requires a larger radius.
- Keep fixed harnesses clear of hinges, pins, actuator rods, wheel/suspension travel, cover edges, and regolith impact zones. Use abrasion sleeve and stand-off clamps where a cable approaches a structure.
- At each moving actuator/bed interface, create a controlled service loop with enough length for the measured path change plus assembly removal allowance. Do not permit the loop to reverse-bend, rub a rod, or enter the hinge pinch zone.
- Terminate shields according to the EMC/bonding plan; do not improvise shield grounding in CAD. Keep RF coax as short as practical and respect its manufacturer bend radius.
- Model connector backshell and tool-access envelopes, not only cable centerlines.

For a two-conductor DC branch, the preliminary voltage-drop relation is

$$\Delta V = 2LI \frac{\rho}{A}, \quad \% \Delta V = 100 \frac{\Delta V}{V_{\text{load}}}, \quad (1)$$

where L is one-way length, I is load current, ρ is conductor resistivity at the design temperature, and A is conductor area. A numerical result is intentionally not reported because conductor material, gauge, length, current, connector resistance, and temperature have not been selected.

10 Preliminary Power and Energy Budget

The only defensible numerical subtotal currently comes from the four sourced COTS packaging candidates. The compute entry is assigned a 30 W continuous operating mode within NVIDIA's published 15–60 W

range; 60 W is used as the candidate peak. The LiDAR peak is the manufacturer's 14 W cold self-heating condition. The 20% factor is an explicit design allocation, not a product specification.

Table 7: Sourced candidate-load subtotal; not the complete rover power budget.

Load	Continuous (W)	Candidate peak (W)	Basis
Jetson AGX Orin-class compute	30.00	60.00	Selected mode and upper value within NVIDIA range [5]
Two Basler ace 2 cameras	6.80	6.80	Two times 3.4 W typical [6]
Livox Mid-360 LiDAR	6.50	14.00	Average and cold self-heating peak [7]
VectorNav VN-100 rugged IMU	0.22	0.22	Manufacturer value [8]
Candidate subtotal	43.52	81.02	Excludes all unselected loads
20% allocation margin	8.70	16.20	Project design allocation
Subtotal with margin	52.22	97.22	Still excludes TBD loads

For the Week 9 trip duration, $t = 3.395$ h, the candidate energy is

$$E_{\text{cand}} = (43.52 \text{ W})(3.395 \text{ h}) = 147.73 \text{ Wh} = 0.148 \text{ kWh}, \quad (2)$$

$$E_{\text{cand},20\%} = 1.20E_{\text{cand}} = 177.28 \text{ Wh} = 0.177 \text{ kWh}. \quad (3)$$

This equals approximately 10.0% of the earlier 1.77 kWh per-trip mission-energy estimate after applying the 20% allocation. The scope of that Week 9 estimate must be checked before adding the values; it may already contain an avionics allowance.

For any selected distribution rail,

$$I_{\text{rail}} = \frac{P_{\text{rail}}}{V_{\text{rail}}\eta_{\text{converter}}}. \quad (4)$$

Numerical rail currents are open because the battery bus, converter efficiency, rail voltages, duty cycles, radio, lamps, actuators, and heaters are not selected. The final budget must separately list continuous, maneuver, transmit, dump, survival-heater, and fault/isolation modes. Actuator and survival-heater peaks may dominate this deck subtotal.

11 Mass, Center of Mass, and Mounting Loads

11.1 Current mass evidence

The current SOLIDWORKS mass-properties window for the `Default` configuration of the rough assembly is reproduced in Figure 4. The recorded values are summarized in Table 8.

This 423.85468 kg value supersedes the earlier 74.2083 kg hidden-body snapshot as the current Week 12 CAD mass record. It remains a modeled-configuration value, not the complete mission dry-mass target. The separate project baselines remain approximately 650 kg dry and 990 kg with the 340 kg payload. Before final release, the active bodies, material assignments, suppressed components, fasteners, harnesses, and internal enclosure hardware must be audited so the CAD snapshot can be reconciled with the mission mass budget.

The two cameras, LiDAR, and rugged IMU have a sourced candidate subtotal

$$m_{\text{sensors}} = 2(0.105) + 0.265 + 0.015 = 0.490 \text{ kg}. \quad (5)$$

Table 8: Current Week 12 SOLIDWORKS mass-properties snapshot.

Property	Recorded value	Converted SI value
Assembly mass	423854.68 g	423.85468 kg
Modeled solid volume	422058841.46 mm ³	0.42205884146 m ³
Surface area	61981198.43 mm ²	61.98119843 m ²
Center of mass, default coordinates	$X = 142.88, Y = 1629.41, Z = 1829.83$ mm	Coordinate-system dependent

Mass properties of rough assem
 Configuration: Default
 Coordinate system: -- default --

Mass = 423854.68 grams

Volume = 422058841.46 cubic millimeters

Surface area = 61981198.43 square millimeters

Center of mass: (millimeters)
 X = 142.88
 Y = 1629.41
 Z = 1829.83

Figure 4: SOLIDWORKS mass properties for the supplied rough-assembly Default configuration. The coordinate values are meaningful only with the displayed default coordinate system and the active component/material state used when the screenshot was captured.

This sourced subtotal shall not be added directly to 423.85468 kg until it is confirmed whether the camera, LiDAR, and IMU placeholder masses are already represented in the active CAD bodies; direct addition could double-count them. The compute assembly, radio, internal electronics, fasteners, connectors, and harness still require selected masses.

For a baseline assembly of mass M_0 and center $\mathbf{r}_0$, the updated center is

$$\mathbf{r}_{\text{new}} = \frac{M_0 \mathbf{r}_0 + \sum_i m_i \mathbf{r}_i}{M_0 + \sum_i m_i}. \quad (6)$$

For the captured configuration, $M_0 = 423.85468$ kg and $\mathbf{r}_0 = (142.88, 1629.41, 1829.83)$ mm in the default coordinate system. Equation 6 can be used after each unrepresented component mass and its coordinates are confirmed. The direction and magnitude of the shift cannot be stated without mapping the default axes to the rover and recording those component coordinates.

11.2 Mounting-load equations

For a component of mass m mounted at height h above the deck and subjected to a selected lateral design acceleration a_d , a first mast/enclosure load screen is

$$V = ma_d, \quad M = ma_d h. \quad (7)$$

If n fasteners share direct shear uniformly, the preliminary fastener shear is V/n . For a plate of thickness t bearing on a fastener of diameter d ,

$$\sigma_{\text{bearing}} = \frac{V}{ntd}. \quad (8)$$

For a rectangular bracket strip of width b and thickness t carrying end moment M , a simple elastic bending screen is

$$\sigma_b = \frac{6M}{bt^2}, \quad \text{FOS} = \frac{\sigma_{\text{allow}}}{\sigma_b}. \quad (9)$$

No numerical bracket FOS is claimed because the design acceleration, component mass, height, actual load-sharing, fastener pattern, bracket dimensions, joint slip, and allowable are unknown. The mast base is the first local bracket study to complete once the antenna and radio are selected.

12 Thermal, Vacuum, Radiation, and Dust Considerations

In lunar vacuum, normal air convection is unavailable. Heat from the computer, converters, drivers, and radio must be conducted through internal straps/baseplates and enclosure feet to a deliberate heat-rejection surface, then rejected primarily by radiation. NASA notes that thermal-vacuum testing is critical for space hardware and supports integrated lunar dust and thermal-vacuum testing [13]. NASA thermal-control guidance also describes thin interface materials between electronics boxes and mounting surfaces as a way to improve conductive heat sinking when adequate contact pressure is maintained [14].

The electronics design should therefore include:

- a conductive mounting plane beneath the central, PDU, and driver enclosures;
- controlled thermal-interface materials and documented clamp pressure;
- radiator surfaces that are not permanently shadowed by the bed and are protected from dust deposition;
- insulation or heaters sized from mission hot/cold cases, not terrestrial ambient ratings;
- temperature sensors on the compute baseplate, converter/PDU hot spot, driver baseplate, radio, both actuators, and a deck/radiator reference;

- vented or vacuum-compatible enclosure construction with no trapped volumes, low-outgassing materials, and suitable lubricants/seals;
- parts screening, derating, radiation analysis, and fault-tolerant architecture appropriate to mission duration and exposure.

The cited COTS products have terrestrial ratings only. For example, the Basler candidate is IP30 and rated -10 – 50°C ; the Livox candidate is IP67 and rated -20 – 55°C ; and the VectorNav rugged IMU is rated -40 – 85°C [6–8]. These ratings do not demonstrate vacuum compatibility, lunar dust tolerance, radiation tolerance, or lunar-night survival.

If the final design were built, dust protection would be applied at every exposed mechanical joint, electrical connection, and cable entry rather than leaving the simplified CAD interfaces open. Electrical connectors would use sealed backshells or sealed bulkhead connectors, cable entries would use sealed glands with strain relief, and unused service ports would remain capped. Hinge pins, actuator clevises, bearings, and sliding interfaces would receive motion-compatible protective covers, vacuum-compatible dust boots, or non-contact labyrinth shields selected for their travel and temperature range. These provisions would reduce direct regolith entry while preserving service access. The final seal geometry and materials are not represented in the current simplified assembly and still require environmental qualification.

13 Interference and Range-of-Motion Verification

The raised configuration in Figure 1 provides useful endpoint evidence, but it is not a full motion proof. The required SOLIDWORKS verification sequence is:

1. Create named configurations: `Bed_Horizontal`, `Bed_Raised`, `Electronics_Installed`, `Sensor_FOV`, `Antenna_Keepout`, and `Harness_Loops`.
2. Model simplified keep-out solids for each lid-removal path, connector backshell, camera FOV, LiDAR FOV, antenna keep-out, cable bend radius, and service loop.
3. Drive the actuator/bed motion through its complete constrained stroke. Use interference detection at both endpoints and a motion-study clearance trace or multiple intermediate positions.
4. Record the minimum distances between the bed/underframe and every enclosure, mast, cable loop, connector, and sensor bracket. Do not accept a visual-only check.
5. Verify that the front cameras and work light are not blocked by the chassis lip or raised bed. Verify rear-sector occlusion separately.
6. Check service states: bed horizontal, bed raised and mechanically supported, enclosure lid removal, connector removal, mast removal, and deck-module removal.

14 Need for Updated Structural Analysis

The Week 11 global structure has large elastic margins in the completed cases, so repeating every global case before assigning actual electronics mass would add little value. The recommended sequence is:

1. Complete the Week 12 BOM, material assignments, enclosure walls, fasteners, mast, brackets, and harness masses.
2. Audit the active bodies and material assignments, then repeat the current 423.85468 kg mass-properties capture for the exact final horizontal and raised configurations.
3. Perform local hand checks or submodel FEA for the antenna mast base, main enclosure feet, PDU/driver brackets, and the front camera/light bar using a documented design acceleration.

Table 9: Current verification status and required closure evidence.

Requirement	Current status	Evidence needed for closure
No payload-floor electronics	PASS	Visible fixed-deck mounting in final CAD
No visible interference at shown raised endpoint	PASS	Final screenshot, Figure 1
Full dump-motion clearance	OPEN	Motion sweep and recorded minimum clearances
Actuator/hinge cable-loop clearance	OPEN	Routed harness bodies in horizontal, raised, and intermediate states
Stereo-camera coverage	OPEN	Second camera/integrated stereo head, lens FOV, baseline and overlap model
LiDAR coverage	OPEN	Mount choice and occlusion model; optional sensor decision
Antenna keep-out and link visibility	OPEN	Selected antenna datasheet, keep-out solid, link directions and link budget
Service/connector access	OPEN	Tool/hand envelopes and removal sequence screenshots
Thermal path and radiator view	OPEN	Thermal model, coating/area, mission hot/cold cases

4. Update the global model if added equipment materially changes the mass distribution, introduces concentrated loads into unsupported deck sheet, or creates local stresses not represented by Week 11.
5. Independently complete the previously open 3207.6 N full-rover load case through representative suspension reactions. This remains a future case, not a Week 11 result.

Local reinforcement is likely required only where a mast or enclosure base lands on unsupported sheet instead of a rail/cross-member. The correct decision depends on actual fastener spacing and load path; the screenshots alone do not justify a doubler thickness.

15 Requirements Assessment

Overall result: preliminary integration PASS with open verification items. The layout is suitable for continuation into detailed electrical and thermal design. It shall not be released as a flight design until all OPEN rows are closed with measured CAD evidence, selected hardware, calculations, and environmental qualification.

16 Risks, Limitations, and Recommended Future Work

1. **Thermal rejection:** the central enclosure may dissipate tens of watts without convection. Define the conductive/radiative path and run mission hot/cold thermal cases before finalizing box locations.
2. **RF uncertainty:** mast height alone does not establish communications performance. Complete the link budget and bed-occlusion study before selecting an antenna.
3. **COTS qualification:** none of the packaging candidates is claimed to meet lunar vacuum, dust, radiation, or thermal-cycling requirements. Develop a screening, derating, redundancy, and test plan.
4. **Perception occlusion:** the current single forward camera placeholder does not prove stereo coverage, and a low LiDAR pedestal may be shadowed by the central enclosure.

Table 10: Week 12 design assessment against defined preliminary requirements.

Requirement	Assessment	Basis / limitation
Electronics on fixed protected structure	PASS	All major modeled items are on the forward fixed deck
Dump path remains unobstructed in shown configuration	PASS	Raised-bed screenshot shows separation
Full motion and pinch-point clearance	OPEN	Motion sweep and harness geometry not supplied
Low and serviceable enclosure arrangement	PASS	Major mass is low; top covers/perimeter connectors are accessible by inspection
Communications hardware provision	PASS	Mast and base housing are modeled; RF performance remains open
No terrestrial GPS assumption	PASS	Baseline uses cameras, odometry, IMU, and optional ranging/lunar PNT
Forward perception provision	PASS	Forward optical pedestal and aperture bar are modeled
Stereo/depth coverage quantified	OPEN	Second camera, optics, and FOV data are not selected
Power distribution and protection provision	PASS	Dedicated PDU/protection and driver enclosure assignments are defined
Complete electrical power budget	OPEN	Only sourced candidate subtotal can be calculated
Harness routing strategy	PASS	Branch separation, clamps, strain relief, service loops and bend-radius rules defined
Harness routing verified in CAD	OPEN	3D cable paths and measured lengths not provided
Built-design dust protection at joints and connections	PASS	Final build specifies sealed connector backshells/bulkheads, sealed cable glands, capped service ports, and protected hinge, actuator, bearing, and sliding interfaces; detailed seal hardware remains to be qualified
Lunar thermal/dust/radiation provisions	OPEN	Design approach defined; qualification analysis and testing not complete
Current CAD mass and center of mass recorded	PASS	Default configuration: 423.85468 kg, with COM (142.88, 1629.41, 1829.83) mm; active-body and material completeness still requires audit
Mounting bracket strength	OPEN	Equations and priority locations defined; inputs/FEA not complete

5. **Harness pinch/abrasion:** moving interfaces are not modeled. Create exact cable paths with minimum bend-radius and service-loop envelopes.
6. **Mass and local loads:** retain the current 423.85468 kg Default-configuration snapshot, but audit its active bodies and material assignments before reconciling it with the 650 kg conceptual dry-mass target or using it for final mast and bracket loads.
7. **Power closure:** select the battery bus, actuator ratings, radio, lamps, heaters, PDU, and converters; then calculate rail currents, protection, connector ratings, losses, voltage drop, and energy by operating mode.
8. **Verification archive:** capture screenshots of each CAD configuration, interference table, minimum clearance, mass properties, component properties, harness length, and any local FEA for the final report revision.

17 Conclusion

The final Week 12 CAD successfully establishes a coherent electronics zone on the rover's fixed forward deck while preserving the visible raised-bed mechanism envelope. The main enclosure, power/protection housing, actuator-driver housing, antenna mast, forward optical sensor, front aperture bar, and service interface form a manufacturable and serviceable preliminary arrangement. The architecture includes every necessary function: protected power entry and conversion, autonomous and safety computing, radio communications, inertial/visual/odometric navigation, mechanism position and limit sensing, hazard perception, lighting, health monitoring, and segmented harness routing.

The strongest design decision is the concentration of heavy electronics low on the fixed deck while leaving only sensors and the antenna exposed. The present design therefore passes the preliminary placement and functional-completeness assessment. The current CAD assembly mass and center of mass have been recorded, and a built version would include sealed electrical interfaces and dust-protected mechanical joints. Quantitative closure remains dependent on component selection, validation of all mass/material assignments, bus and actuator data, link requirements, thermal cases, stereo/FOV geometry, dust-seal qualification, and swept-volume harness clearance. Those open items are explicitly preserved rather than replaced by invented values.

18 References

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